

Weaknesses of Sboxes based on 3-round Feistel Structures

Merlin Fruchon

Inria and DGA, France

WCC, 2026-06-09



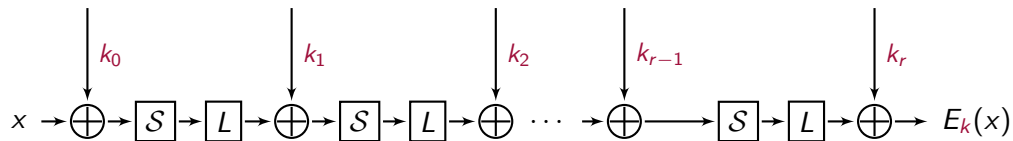
Outline

- 1 Motivation and context
- 2 Commutation with 3-round Feistel Sboxes
- 3 Nonlinear invariants of 3-round Feistel Sboxes
- 4 Perfect sets of differences for 3-round Feistel Sboxes
- 5 Conclusion

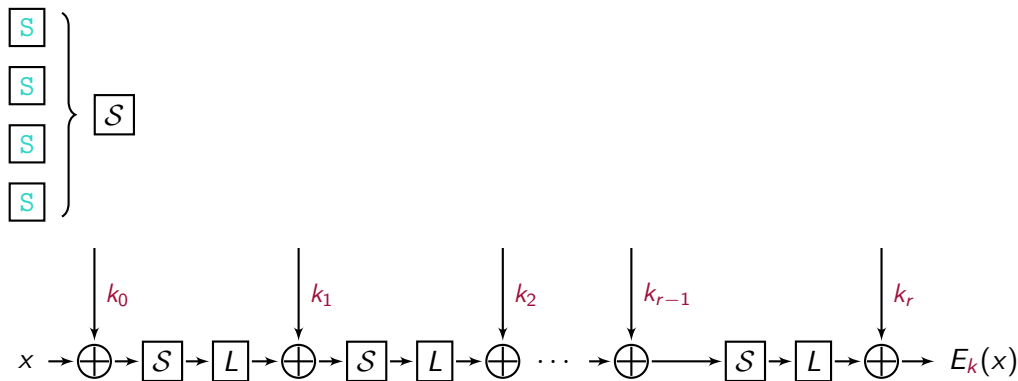
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SPN ciphers



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Resistance to statistical attacks: the key role of the Sbox

Linear Cryptanalysis [Mat94]

Goal of the **attacker**: find (α, β) such that $\alpha \cdot x + \beta \cdot E_k(x) = \varepsilon$ holds with high probability for some $\varepsilon \in \mathbb{F}_2$.

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- $\lambda(\mathbf{s}) := \max_{\alpha, \beta \neq (0,0)} \left| \sum_x (-1)^{\alpha \cdot x + \beta \cdot \mathbf{s}(x)} \right|$
- $\delta(\mathbf{s}) := \max_{a, b \neq (0,0)} \# \{x \mid \mathbf{s}(x) \oplus \mathbf{s}(x \oplus a) = b\}$
- Goal of the **designer**: minimize $\lambda(\mathbf{s})$ and $\delta(\mathbf{s})$.

Construction of Sboxes with "large" number of variables

8-bit Sboxes

- For 8-bit bijections S the optimal values for $\lambda(S)$ and $\delta(S)$ are unknown.
- The current best are achieved by $S : x \mapsto x^{254}$.

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Use of constructions based on smaller Sboxes

- Low implementation cost.
- Well-understood cryptographic properties.

Feistel networks (1971)

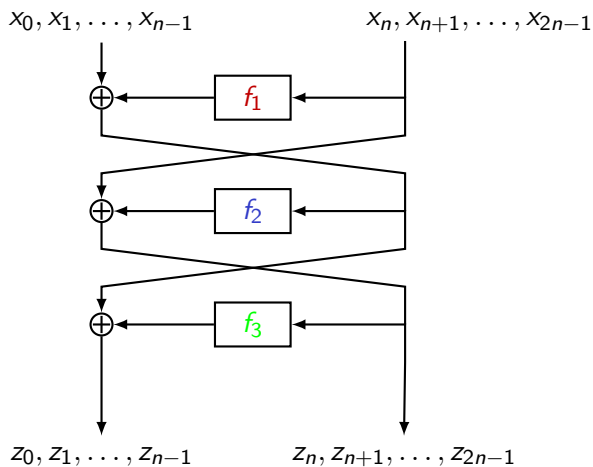


Figure: Feistel(f_1, f_2, f_3) where $f_1, f_2, f_3 : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$

Feistel-based Sboxes: Properties

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f_2 and f_4 are **bijective** 4-bit Sboxes and f_1, f_3 are **APN**.

Lightweight ciphers are vulnerable to new type of attacks

New weak-key attacks against (some) lightweight ciphers

- Commutative Cryptanalysis [BFL⁺23]
- Nonlinear Invariants [TLS16, TLS19, Bey18]
- **Fixed-key** Differential Cryptanalysis [CF26].

Surprisingly, these attacks often apply to the **same** ciphers (e.g SCREAM and MIDORI64).

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Our work: give **sufficient conditions** on $f_1, f_2, f_3 \implies S = \text{Feistel}(f_1, f_2, f_3)$ weak.

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(Affine) Commutative Cryptanalysis [Wag04, BFL⁺23]

Goal of the **attacker**: find a pair of (affine) functions $(\mathcal{B}, \mathcal{C})$ such that $\mathcal{B} \circ E_k = E_k \circ \mathcal{C}$.

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Commutative trail

If $E_k = \mathcal{R}_r \circ \dots \circ \mathcal{R}_2 \circ \mathcal{R}_1$, an attacker may look for $\mathcal{B} = \mathcal{B}_0, \dots, \mathcal{B}_r = \mathcal{C}$ such that

$$\mathcal{R}_i \circ \mathcal{B}_{i-1} = \mathcal{B}_i \circ \mathcal{R}_i.$$

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Main difficulty: find $(\mathcal{B}, \mathcal{C})$ s.t. $\mathcal{B} \circ \mathcal{S} = \mathcal{S} \circ \mathcal{C}$.

$\mathcal{B}$ and $\mathcal{C}$ affine $\iff \mathcal{S}$ is affinely self-equivalent.

A sufficient condition for commuting with 3-round Feistel Sboxes

Derivative

$$D_a f : x \mapsto f(x) \oplus f(x \oplus a).$$

Proposition

Let $S := \text{Feistel}(f_1, f_2, f_3)$ and

$$\begin{aligned} \phi_1 &: (x, y) \mapsto (x + D_a f_1, y + a) \\ \phi_3 &: (x, y) \mapsto (x + D_a f_3, y + a) \end{aligned} .$$

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- For iSCREAM, we found **15** non-affine commuting functions. How to exploit them?

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Nonlinear invariant attacks [TLS16, TLS19, Bey18]

Goal of the **attacker**: find a set $\mathcal{X}$ such that $E_k(\mathcal{X}) = \mathcal{X}$ or $E_k(\mathcal{X}) = \mathbb{F}_2^n \setminus \mathcal{X}$.
Equivalently, find a Boolean function g such that $g \circ E_k + g$ is constant.

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$\implies$ Search for **quadratic** invariants of S .

A sufficient condition for 3-round Feistel Sboxes to have quadratic invariants

Proposition

Let $S := \text{Feistel}(f_1, f_2, f_3)$ and assume that there exist α, ε such that $\alpha \cdot f_1 = \alpha \cdot f_3 + \varepsilon$ is **quadratic**.

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- Using that $0x2 \cdot f_1 = 0x2 \cdot f_3 + 1$ is **quadratic**, we retrieve the only **quadratic** invariant for SCREAM's Sbox found by Todo, Leander and Sasaki.

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- Using that $0x1 \cdot f_4$ and $0x4 \cdot f_4$ are **quadratic**, we retrieve **2 out of the 6** independent **quadratic** invariants found by the authors.

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Perfect sets of differences (adapted from [CF26])

- Valid inputs: $\text{XDDT}_{\mathbf{s}}(\mathbf{a}, \mathbf{b}) := \{x \in \mathbb{F}_2^n \mid \mathbf{s}(x) + \mathbf{s}(x + \mathbf{a}) = \mathbf{b}\}$
- Valid outputs: $\text{YDDT}_{\mathbf{s}}(\mathbf{a}, \mathbf{b}) := \{\mathbf{s}(x) \in \mathbb{F}_2^n \mid \mathbf{s}(x) + \mathbf{s}(x + \mathbf{a}) = \mathbf{b}\}$
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Definition when $\delta(\mathbf{S}) = 4$

$\mathcal{A} \subset \mathbb{F}_2^n$ is a **good set of differences** for $\mathbf{S}$ if:

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- 2 There exists a **linear space** $\mathbf{E}$, s.t. for all $\mathbf{a} \in \mathcal{A}$,

$$\biguplus_{\mathbf{b} \in \mathcal{A}} \text{XDDT}_{\mathbf{S}}(\mathbf{a}, \mathbf{b}) \text{ is an additive coset of } \mathbf{E}.$$

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- The linear parts of all $\text{YDDT}_{\mathbf{S}}(a, b)$ are included in E .

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If two more conditions on the offsets of $\text{XDDT}_{\mathbf{S}}(a, b)$ are satisfied, $\mathcal{A}$ is a **perfect set of differences**.

A perfect set of differences for MIDORI64 Sbox

For all differentials $(a, b) \in \mathcal{A} := \{0xa, 0xf\}$, $\text{DDT}_{\text{S}}(a, b) = 4$.

$$\begin{array}{llll}
 \text{XDDT}_{\text{S}}(0xa, 0xa) & = 0x3 + \langle 0x7, 0xa \rangle & \text{and} & \text{YDDT}_{\text{S}}(0xa, 0xa) = 0x3 + \langle 0x7, 0xa \rangle \\
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 \text{XDDT}_s(0xf, 0xf) & = 0x2 + \langle 0xa, 0xf \rangle & \text{and} & \text{YDDT}_s(0xf, 0xf) = 0xd + \langle 0xa, 0xf \rangle
 \end{array}$$

$E := \langle 0x7, 0xa, 0xf \rangle$ satisfies:

- $\forall a \in \mathcal{A}, \biguplus_{b \in \mathcal{A}} \text{XDDT}_s(a, b)$ is a coset of E
- The linear parts of all $\text{YDDT}_s(a, b)$ are included in E .
- The conditions on the offsets are satisfied.

Proposition (adapted from [CF26])

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Theorem

Let $\mathcal{S} := \text{Feistel}(f_1, f_2, f_3)$ and assume that:

- f_1 and f_3 are **APN**, f_2 is **invertible**.
- There exists b such that $D_b f_1 = D_b f_3$ is **affine**.

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Applications

Allows us to retrieve the **perfect set of differences** for SCREAM found in [CF26].

This section

- 1 Motivation and context
- 2 Commutation with 3-round Feistel Sboxes
- 3 Nonlinear invariants of 3-round Feistel Sboxes
- 4 Perfect sets of differences for 3-round Feistel Sboxes
- 5 Conclusion**

Conclusion

Sbox \ Property	Cipher	λ, δ	Affine commutant	Quadratic invariants	Fixed-key differentials
Feistel(f_1, f_2, f_3)	SCREAM	64, 8	✓	✓	✓
Feistel(f_4, f_4, f_4)	iSCREAM	64, 16	✗	✓	✗
Feistel(f_1, f_1, f_3)		128, 42	✓	✓	✗
Feistel(f_5, f_1, f_6)		128, 36	✓	✗	✗
Feistel(f_5, f_2, f_6)		64, 8	✓	✗	✓
Feistel(f_1, f_4, f_6)		64, 8	✗	✗	✗

f_1, f_3, f_5 and f_6 are **APN**, f_2 and f_4 are **bijections**.



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